

Lecture 5: Speaker Notes

Statistical Analysis Plans and Project 1

Dr. Ronald G. Thomas

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Slide: Title

- Welcome students to Lecture 5
 - Today marks the transition from foundational skills to applied work
 - We covered reproducibility, renv/Docker, and Git/GitHub
 - Now we put those skills into practice with Project 1
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Slide: Course Information

- Remind students of office hours availability
 - TA Man Luo available for project questions
 - Encourage early engagement if struggling with concepts
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Slide: Today's Agenda

- Two main topics: (1) analysis plans, (2) Palmer Penguins data
 - Analysis plans are the “contract” for your work
 - Penguins dataset is your first real collaborative project
 - **Discussion prompt:** Ask about their experiences starting analyses without clear plans—what went wrong?
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Slide: What is a Statistical Analysis Plan?

- Emphasize PRE-SPECIFICATION—written before seeing data
 - SAPs prevent “data dredging” and p-hacking
 - In clinical trials, SAPs are legally required documents
 - Even for exploratory work, planning improves quality
 - **Key point:** The plan documents your intentions, not your results
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Slide: Why Analysis Plans Matter

- Reference the Slade et al. survey: 72.3% rate as essential
 - This is a core professional skill, not just academic exercise
 - FDA requires SAPs for drug approval submissions
 - Journals increasingly require pre-registration
 - **Discussion prompt:** How does a plan help when results disappoint the PI? (Answer: protects the statistician)
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Slide: Elements of an Analysis Plan (1)

- Introduction sets context—why does this matter?
 - Research questions must be SPECIFIC and ANSWERABLE
 - Data description: be precise about what you have
 - **Common mistake:** Vague questions like “explore the data”
 - Good questions specify outcome, predictors, and population
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Slide: Elements of an Analysis Plan (2)

- Methods section is the heart of the plan
 - Specify exact statistical procedures
 - Pre-specify significance level and any adjustments
 - Results presentation: table shells force clear thinking
 - **Teaching point:** If you can't fill a table shell, you're missing something in your methods
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Slide: Meet the Palmer Penguins

- Dr. Kristen Gorman collected data at Palmer Station
 - Three species: Adelie, Chinstrap, Gentoo
 - This replaces the iris dataset for modern teaching
 - Real ecological context: Antarctic climate change
 - **Engagement:** Show enthusiasm—penguins are charismatic!
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Slide: Dataset Structure

- Walk through each variable briefly
 - Emphasize units (mm, g) for morphometric measurements
 - Note the factor variables: species, island, sex
 - Year variable allows temporal analysis (but not primary focus)
 - **Key point:** Body mass is typical outcome; species often modifies relationships
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Slide: Species Characteristics

- Gentoo are clearly largest—good visual separation
 - Adelie and Chinstrap overlap substantially
 - Unequal sample sizes: Chinstrap smallest group
 - Sexual dimorphism exists but not shown in this plot
 - **Discussion:** What analytical challenges do unequal groups create?
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Slide: Key Relationships

- Flipper length is strongest predictor ($r = 0.87$)
 - Bill depth shows **NEGATIVE** correlation overall
 - This is Simpson's paradox—crucial teaching moment
 - Within species, bill depth is positively correlated
 - **Spend time here:** Simpson's paradox is counterintuitive but essential to understand
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Slide: Data Quality Considerations

- Only 11 missing values (3.2%)—relatively clean data
 - Complete case analysis is reasonable here
 - Discuss when imputation would be preferred
 - Mention potential outliers in morphometric data
 - Year effects: 3 years of data, possible confounding
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Slide: Project 1 Analysis Plan Overview

- Present sample research question as MODEL
 - Note it specifies outcome, predictors, and comparison
 - Objectives are numbered and actionable
 - **Critique the question:** Could be more specific about which models, what “vary by species” means exactly
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Slide: Sample Plan - Data Description

- Three key elements: source, population, sample size
 - Citation is essential for reproducibility
 - “Expected n = 333” shows you've thought about missingness
 - **Teaching point:** This section should let someone else obtain identical data
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Slide: Sample Plan - Variables

- Distinguish outcome, predictors, effect modifiers
 - Effect modifiers suggest interaction terms
 - Predictors have direct effects
 - **Common confusion:** When is species a predictor vs. modifier? Depends on research question
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Slide: Sample Plan - Statistical Methods

- Sequential model building is pedagogically valuable
 - Shows contribution of each component
 - Model 1: baseline, Model 2: multiple predictors, etc.
 - Comparison criteria specified in advance
 - **Key point:** Don't just report the “best” model—show the progression
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Slide: Sample Plan - Model Evaluation

- R-squared for variance explained
 - RMSE in original units (grams)—interpretable
 - AIC/BIC for model selection
 - Diagnostics: residual plots are fundamental
 - **Emphasis:** Always check assumptions before interpreting
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Slide: Sample Plan - Results Presentation

- Table shells are REQUIRED in clinical trials
 - Forces you to think through exactly what you'll report
 - 4-6 figures is appropriate for this project
 - Each figure should answer a specific question
 - **Activity idea:** Have students sketch a table shell
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Slide: Sample Plan - Sensitivity Analyses

- Robustness checks test if conclusions depend on choices
 - Compare complete case to imputation
 - Check influence of outliers
 - Stratified vs. pooled models
 - Limitations section shows intellectual honesty
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Slide: Key Takeaways

- Write plan BEFORE analyzing
 - Include all four elements: background, data, methods, results
 - Be specific enough for someone else to execute
 - Palmer Penguins: 344 obs, 3 species, $r = 0.87$ for flipper
 - Simpson's paradox is key insight
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Slide: Assignment - Project 1 Plan

- Due Thursday January 22—in-class presentation
 - 15-20 minutes per team
 - ALL members must participate in presentation
 - Submit slides to Canvas BEFORE class
 - Template provided: `analysis_plan_template.qmd`
 - **Encourage:** Start today, don't wait until Wednesday night
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Slide: References

- Point students to the annotated bibliography in lec05
 - Gorman et al. for biological context
 - Horst et al. for dataset documentation
 - Slade et al. for professional importance of SAPs
 - ICH E9 for regulatory perspective (optional reading)
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General Teaching Notes

Pacing:

- Part I (Analysis Plans): ~20 minutes
- Part II (Palmer Penguins): ~15 minutes
- Part III (Sample Plan): ~20 minutes
- Q&A and Assignment: ~5 minutes

Materials to have ready:

- Rendered slides (`slides.html`)
- Analysis plan template (`analysis_plan_template.qmd`)
- Annotated bibliography for reference
- `palmerpenguins_rjournal.pdf` for deeper questions

Anticipated questions:

1. “How long should the plan be?” → Quality over length; 5-8 pages typical for this project
2. “Can we change the plan later?” → Yes, but document changes and justify them
3. “What if our results don't match hypotheses?” → That's fine; report what you find, not what you hoped for
4. “How specific should model equations be?” → Specific enough to code directly from the plan

Common pitfalls to warn about:

- Starting analysis before finishing plan
- Vague research questions
- Forgetting to specify handling of missing data
- Not pre-specifying significance level
- Table shells that don't match methods